HR ATTRITION ANALYSIS & PREDICTION (END-TO-END PROJECT)

Presented By Pallab Kanti Lahiri

PROJECT OVERVIEW

- This project analyzes employee data to identify factors affecting attrition and builds a machine learning model to predict whether an employee is likely to leave the company.

OBJECTIVES

- Analyze employee attrition trends
- Identify key factors influencing attrition
- Build a predictive model using Machine Learning
- Provide actionable business insights

DATA CLEANING

- Removed irrelevant columns (EmployeeCount, StandardHours, Over18)
- Checked for missing values and duplicates
- Converted target variable (Attrition: Yes → 1, No → 0)
- Applied One-Hot Encoding using pandas

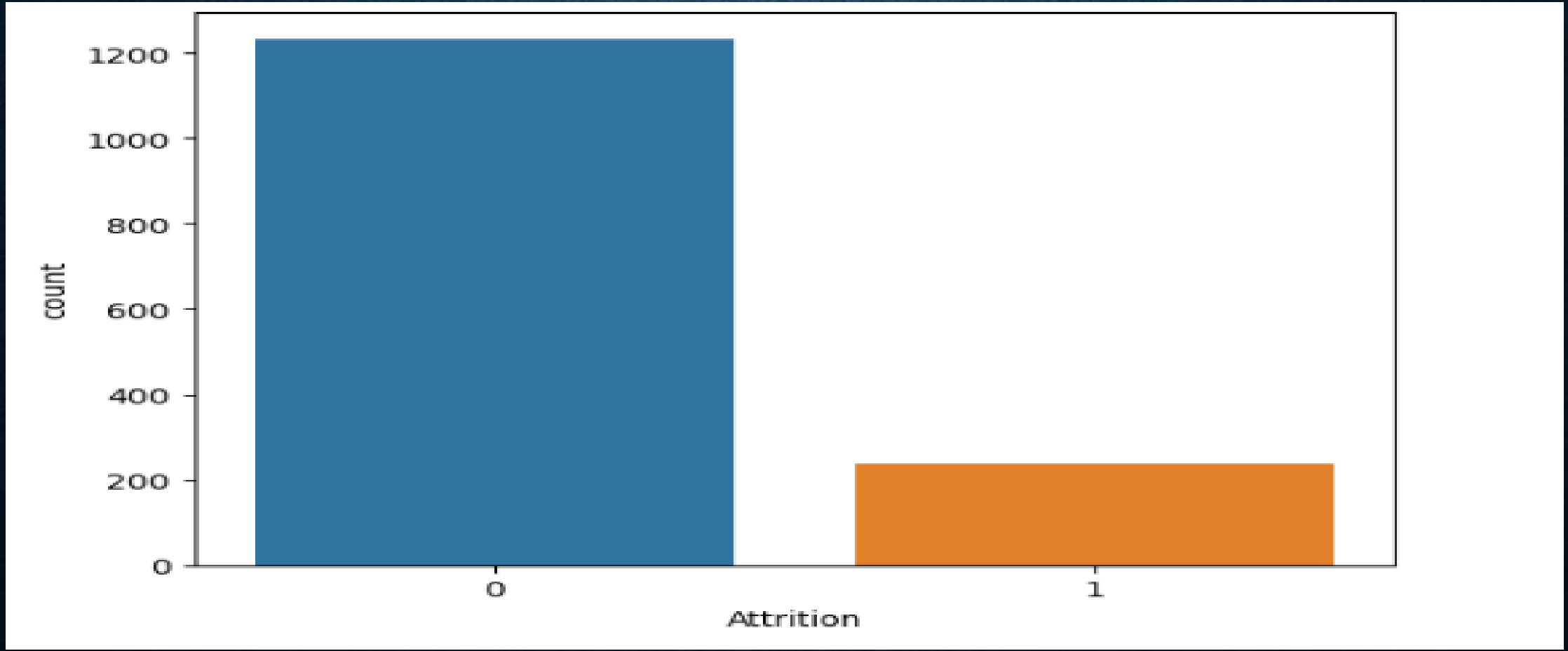
DATA CLEANING

Data Cleaning

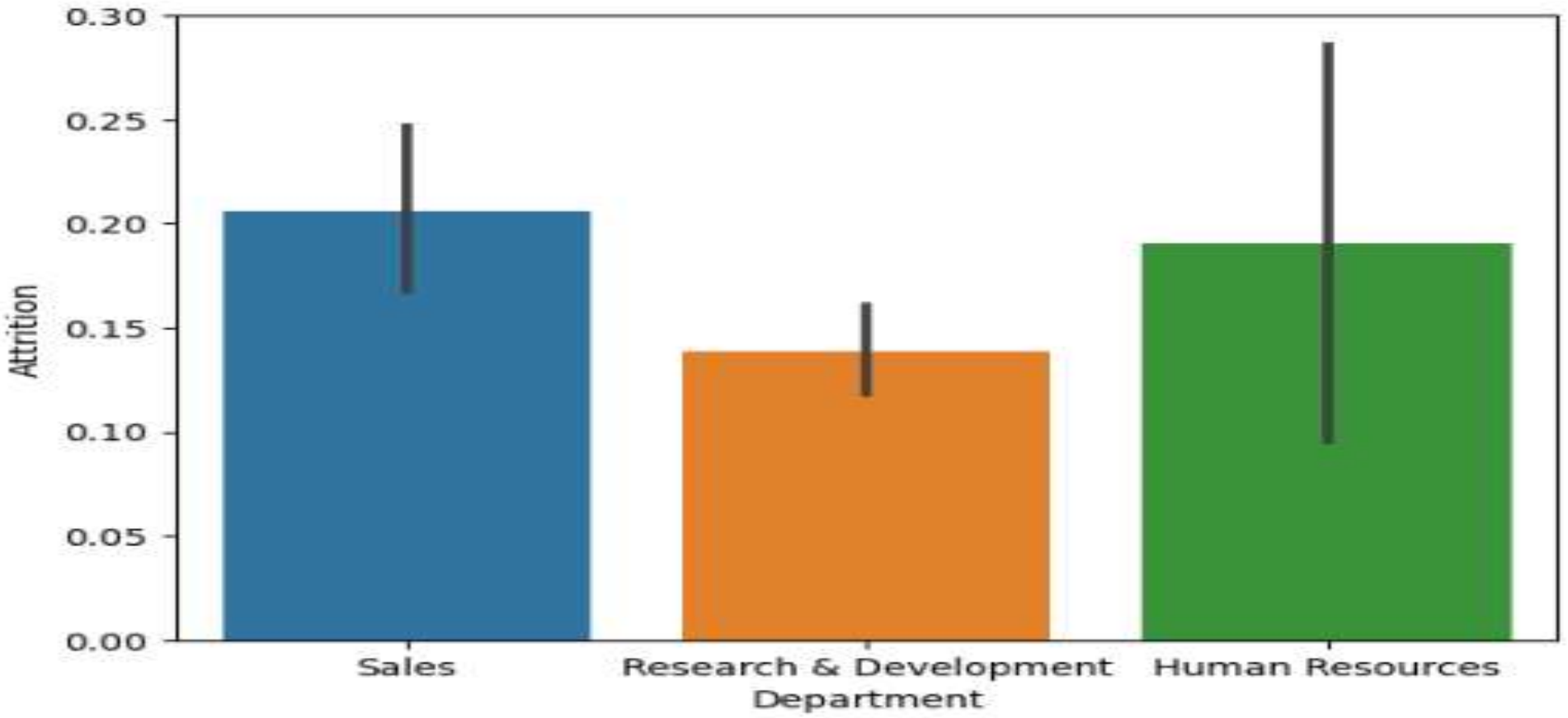
```
df.isnull().sum()  # Check Missing Values
```

```
Age                                 0
Attrition                           0
BusinessTravel                      0
DailyRate                           0
Department                          0
DistanceFromHome                    0
Education                           0
EducationField                      0
EmployeeCount                       0
EmployeeNumber                      0
EnvironmentSatisfaction             0
Gender                              0
HourlyRate                          0
JobInvolvement                      0
JobLevel                            0
JobRole                             0
JobSatisfaction                     0
MaritalStatus                       0
MonthlyIncome                       0
MonthlyRate                         0
NumCompaniesWorked                  0
Over18                              0
OverTime                            0
PercentSalaryHike                   0
PerformanceRating                   0
RelationshipSatisfaction            0
StandardHours                       0
StockOptionLevel                    0
TotalWorkingYears                   0
TrainingTimesLastYear               0
WorkLifeBalance                     0
YearsAtCompany                      0
YearsInCurrentRole                  0
YearsSinceLastPromotion             0
YearsWithCurrManager                0
dtype: int64
```

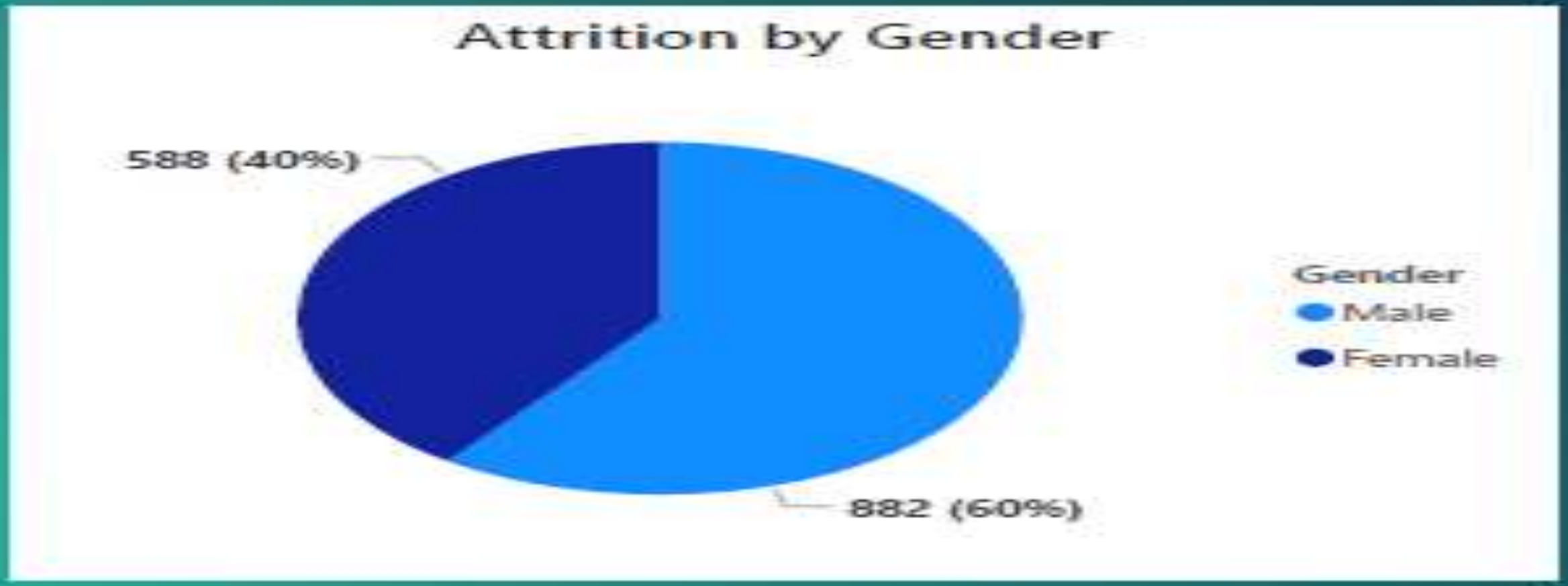
EXPLORATORY DATA ANALYSIS (EDA)
ATTRITION DISTRIBUTION ANALYSIS



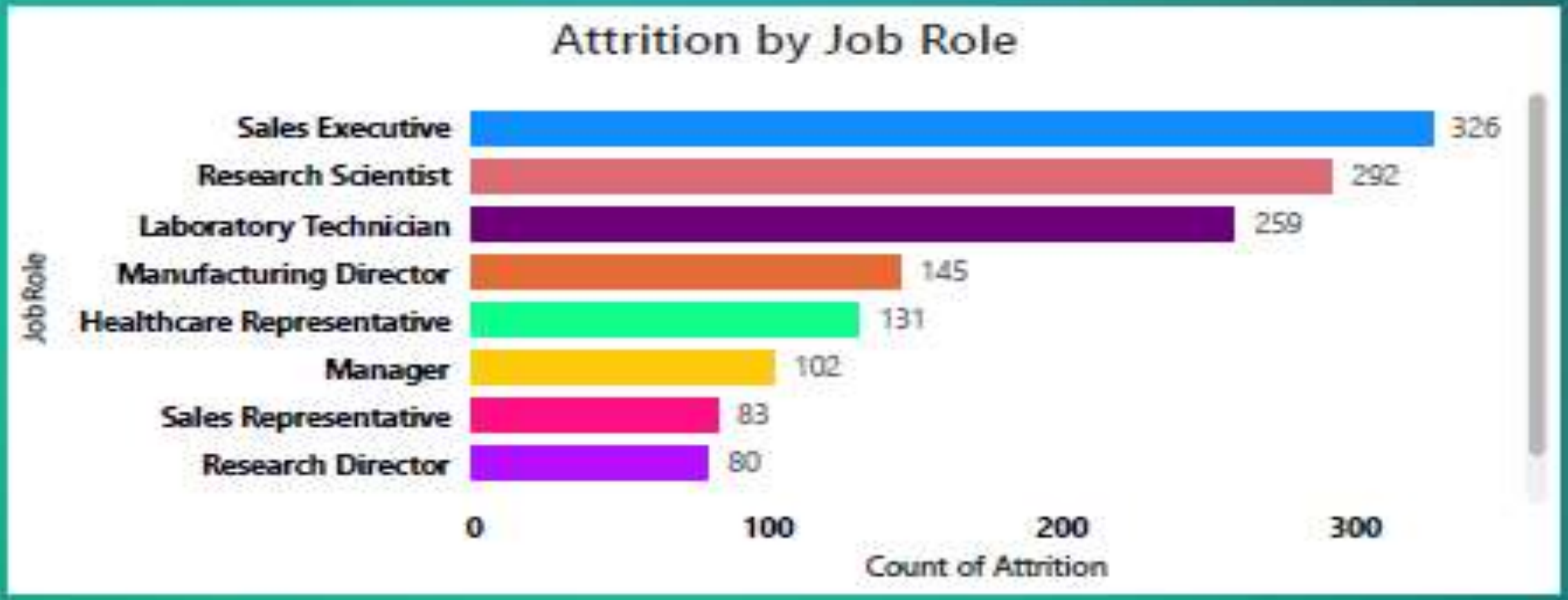
EXPLORATORY DATA ANALYSIS (EDA)
ATTRITION BY DEPARTMENT ANALYSIS



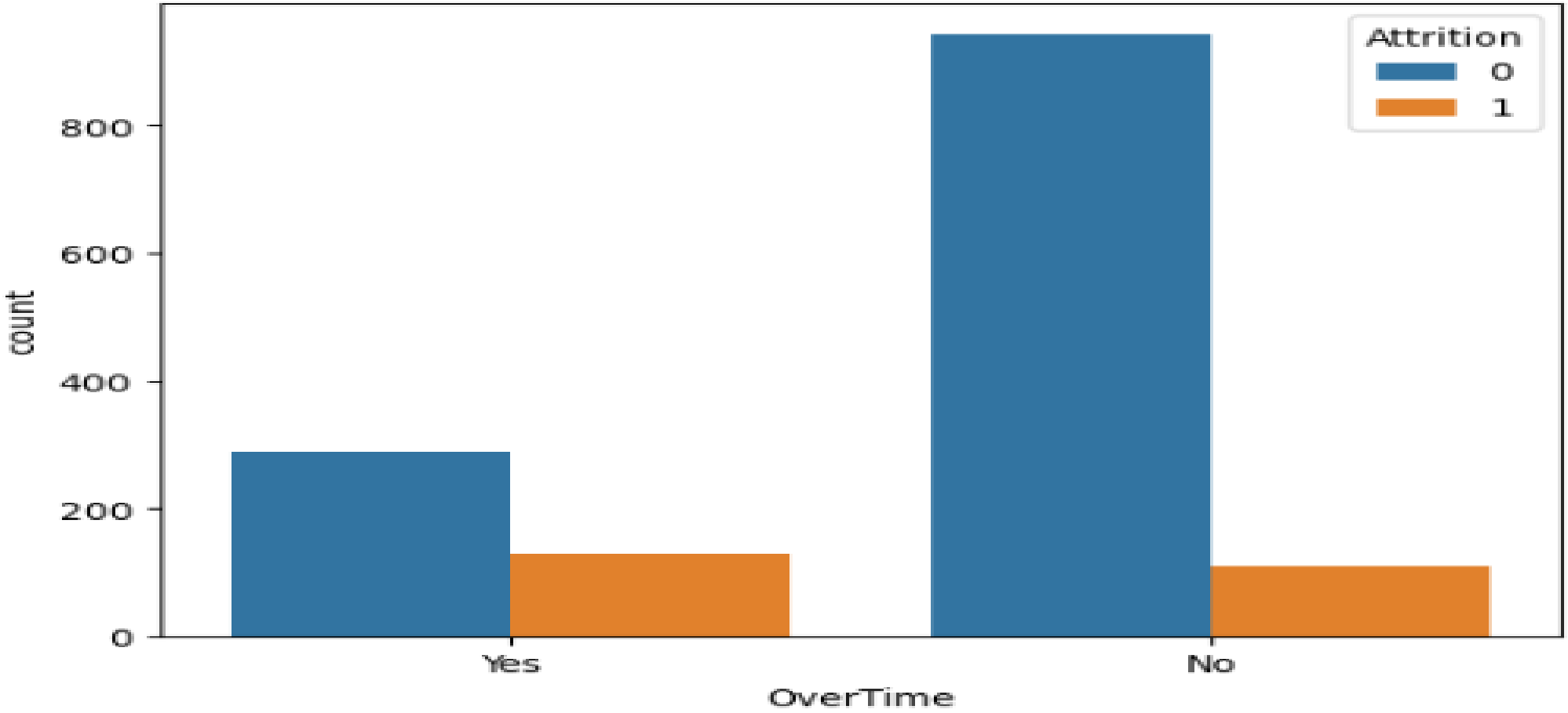
EXPLORATORY DATA ANALYSIS (EDA)
ATTRITION BY GENDER ANALYSIS



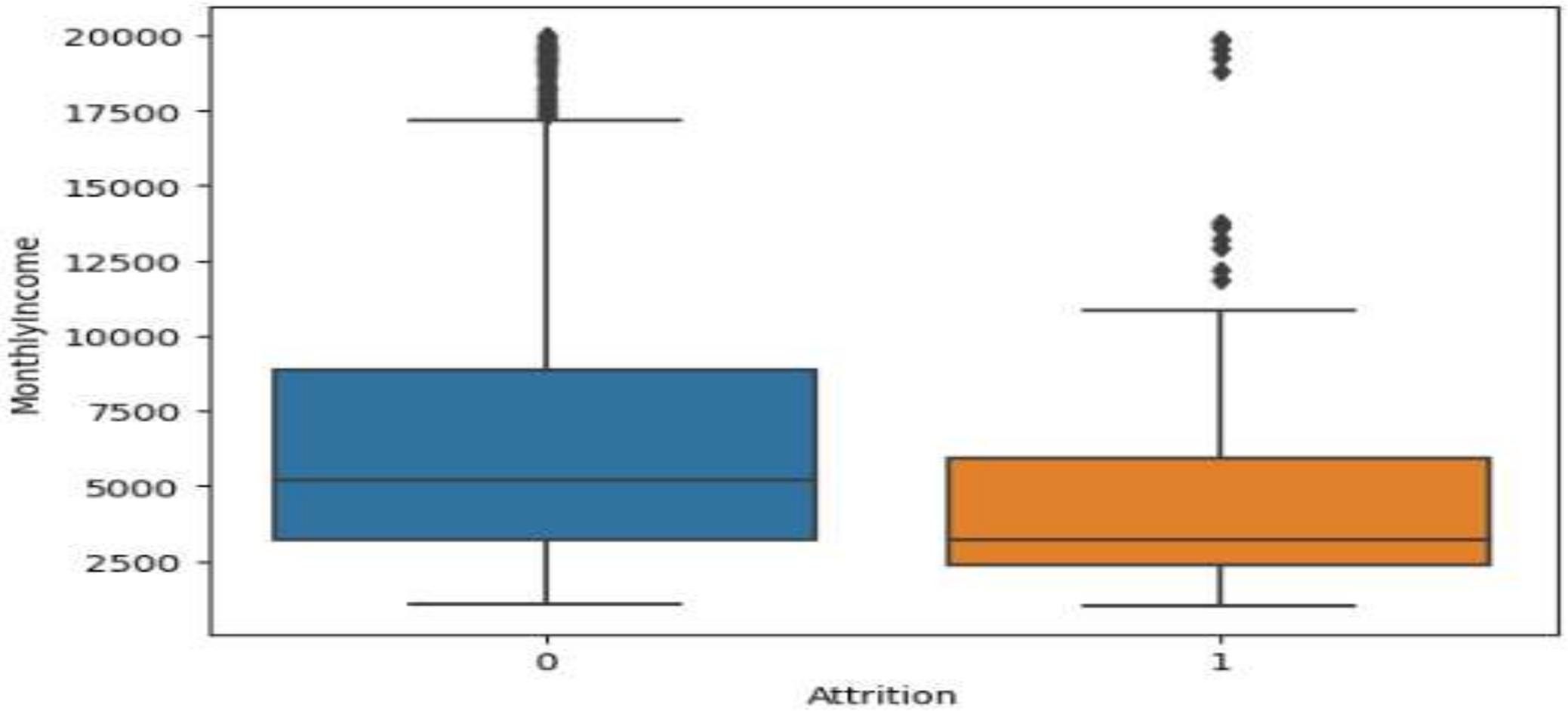
EXPLORATORY DATA ANALYSIS (EDA)
ATTRITION BY JOB ROLE ANALYSIS



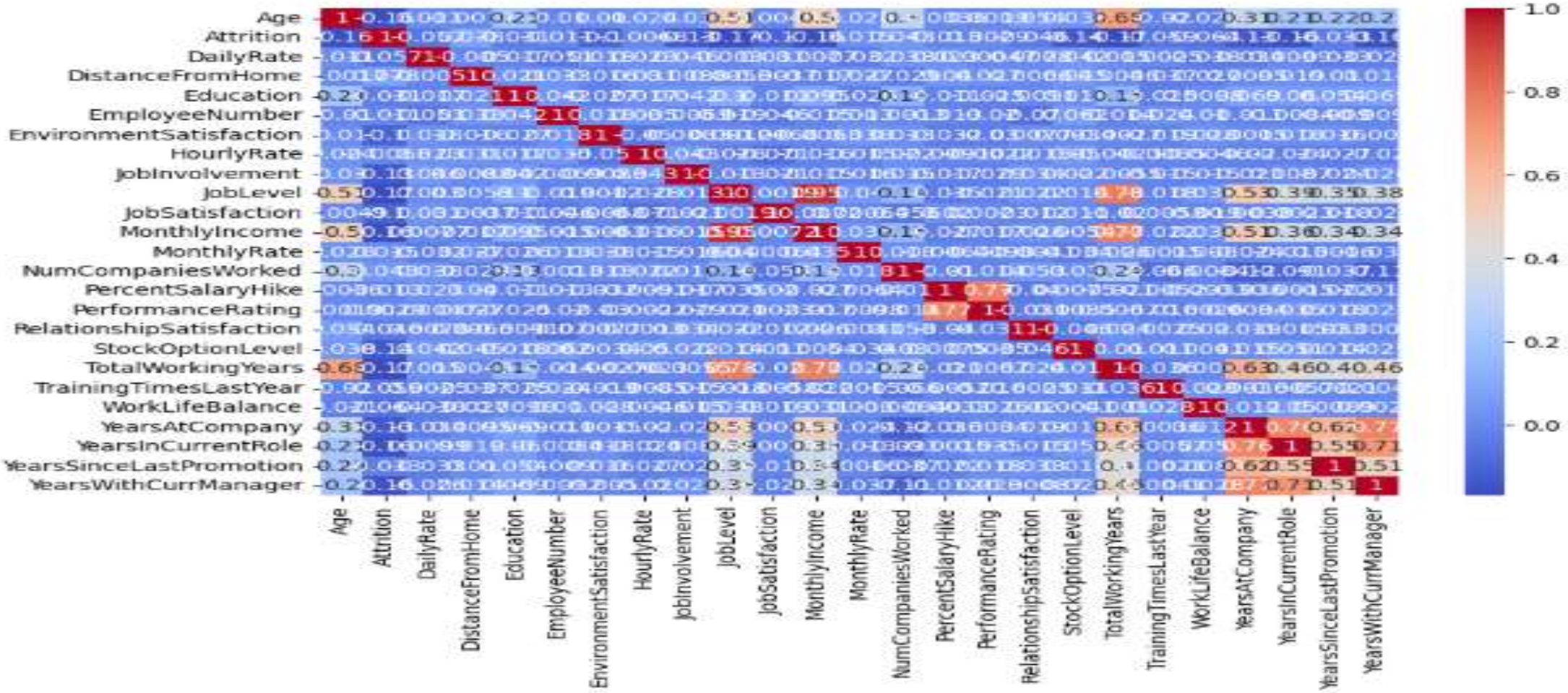
EXPLORATORY DATA ANALYSIS (EDA)
ATTRITION BY OVERTIME ANALYSIS



EXPLORATORY DATA ANALYSIS (EDA)
ATTRITION BY SALARY ANALYSIS



EXPLORATORY DATA ANALYSIS (EDA)
CORRELATION HEATMAP ANALYSIS



FEATURE ENGINEERING
CREATE AGE GROUP

nesLastYear	WorkLifeBalance	YearsAtCompany	YearsInCurrentRole	YearsSinceLastPromotion	YearsWithCurrManager	AgeGroup
0	1	6	4	0	5	36-45
3	3	10	7	1	7	46-55
3	3	0	0	0	0	36-45
3	3	8	7	3	0	26-35
3	3	2	2	2	2	26-35

FEATURE ENGINEERING
CREATE TENURE GROUP

VorkLifeBalance	YearsAtCompany	YearsInCurrentRole	YearsSinceLastPromotion	YearsWithCurrManager	AgeGroup	TenureGroup
1	6	4	0	5	36-45	6-10
3	10	7	1	7	46-55	6-10
3	0	0	0	0	36-45	NaN
3	8	7	3	0	26-35	6-10
3	2	2	2	2	26-35	0-2

ADVANCE ANALYSIS

Advanced Analysis

```
# Attrition Rate
attrition_rate= df['Attrition'].mean()*100
```

```
# High Risk Employees

high_risk = df[(df['OverTime'] == 'Yes') & (df['JobSatisfaction']<=2) & (df['YearsSinceLastPromotion']>3)]
```

```
# Group Analysis

df.groupby('JobRole').agg({'Attrition':'mean', 'MonthlyIncome': 'mean'}).sort_values(by='Attrition', ascending=False)
```

JobRole	Attrition	MonthlyIncome
Sales Representative	0.397590	2626.000000
Laboratory Technician	0.239382	3237.169884
Human Resources	0.230769	4235.750000
Sales Executive	0.174847	6924.279141
Research Scientist	0.160959	3239.972603
Manufacturing Director	0.068966	7295.137931
Healthcare Representative	0.068702	7528.763359
Manager	0.049020	17181.676471
Research Director	0.025000	16033.550000

MACHINE LEARNING PREDICTION MODEL PREDICT WHETHER AN EMPLOYEE WILL LEAVE (ATTRITION=1) OR STAY(0)

- Model Used: 1. Logistic Regression. 2. Random Forest Classifier.
- Data Preprocessing: 1. Train-Test-Split. 2. Feature Scaling
- Evaluation Metrics: 1. Accuracy. 2. Confusion Matrix. 3. Classification Report

FEATURE IMPORTANCE

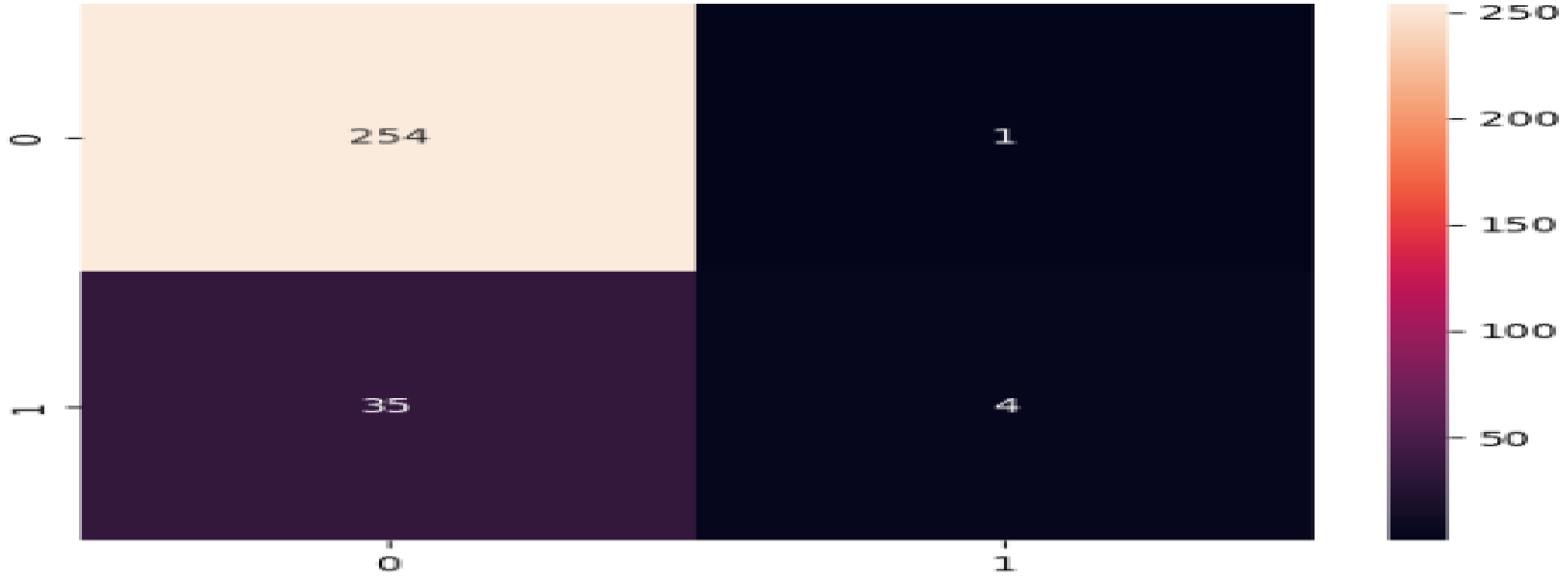
Feature Importance

```
importances = rf.feature_importances_

feature_importance = pd.DataFrame({
    'Feature': X.columns,
    'Importance': importances
}).sort_values(by='Importance', ascending=False)

print(feature_importance.head(10))
```

```
                  Feature  Importance
10          MonthlyIncome    0.074927
44           OverTime_Yes    0.060696
0                     Age    0.049015
11            MonthlyRate    0.044270
17    TotalWorkingYears    0.043394
1               DailyRate    0.042469
4          EmployeeNumber    0.041531
6               HourlyRate    0.040478
20        YearsAtCompany    0.040253
2       DistanceFromHome    0.037208
```

MACHINE LEARNING PREDICTION MODEL PREDICT WHETHER AN EMPLOYEE WILL LEAVE (ATTRITION=1) OR STAY(0) MODEL EVALUATION

```
# Confusion Matrix

sns.heatmap(confusion_matrix(y_test, y_pred_rf), annot=True, fmt='d')
plt.show()
```



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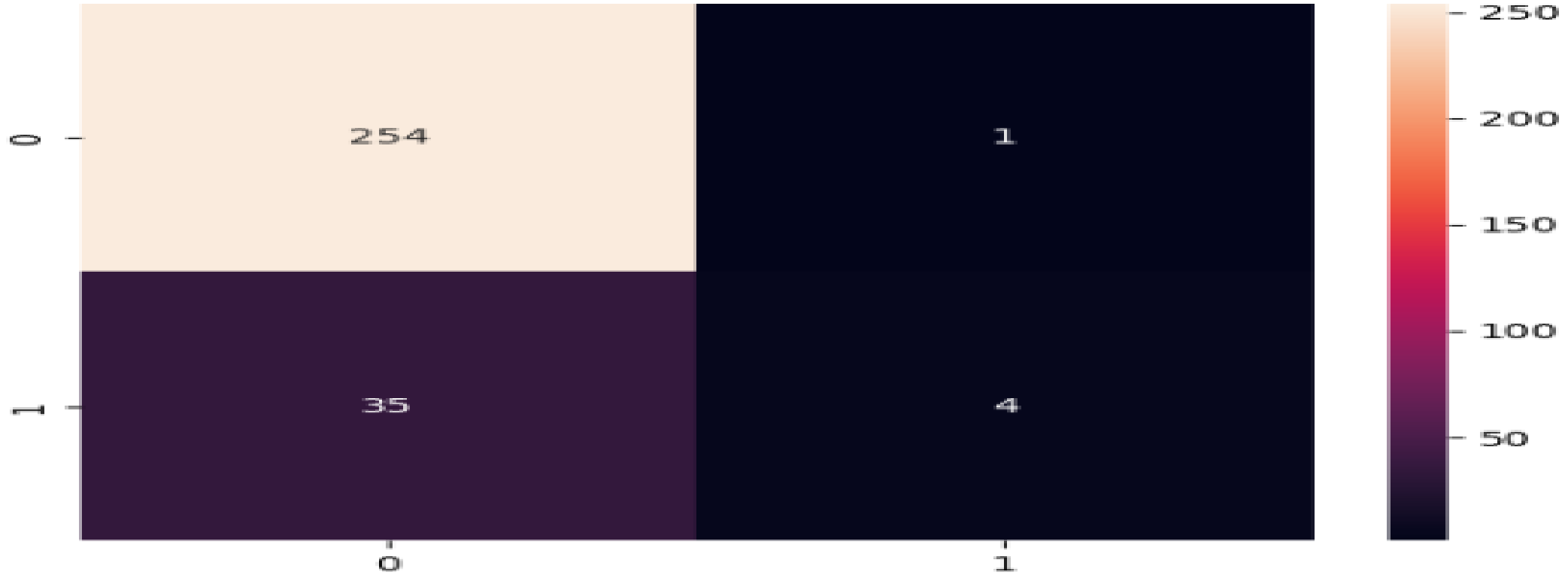


KEY INSIGHTS OF PYTHON ANALYSIS

- Employees working overtime are more likely to leave
- Low job satisfaction increases attrition risk
- Employees with fewer years at company leave more
- Salary has moderate impact on attrition

KEY INSIGHTS OF POWER BI ANALYSIS

- I recently worked on an HR Analytics Dashboard using Power BI to uncover key workforce trends and employee attrition patterns. Here are some impactful insights from the analysis

KEY INSIGHTS OF POWER BI ANALYSIS

•Key Highlights:

✅ Total Employees: ~2,000

✅ Attrition Count: 237

✅ Attrition Rate: 16.12%

✅ Average Salary: 6.5K

✅ Average Age: 36.9 years

KEY INSIGHTS OF POWER BI ANALYSIS

- 📌 **Department-wise Attrition**

  → 🔹 **Highest attrition observed in Research & Development, followed by Sales**

  → 🔹 **HR department shows minimal attrition**

- 📌 **Gender Distribution**

  → 🔹 **Male employees contribute 60% of attrition**

  → 🔹 **Female employees account for 40%**

KEY INSIGHTS OF POWER BI ANALYSIS

- 📌 Job Role Analysis
  → Sales Executives, Research Scientists, and Lab Technicians show the highest attrition
  → Leadership roles like Managers and Directors have relatively lower attrition
- 📌 Overtime Impact
  → Surprisingly, overtime is NOT a major factor driving attrition

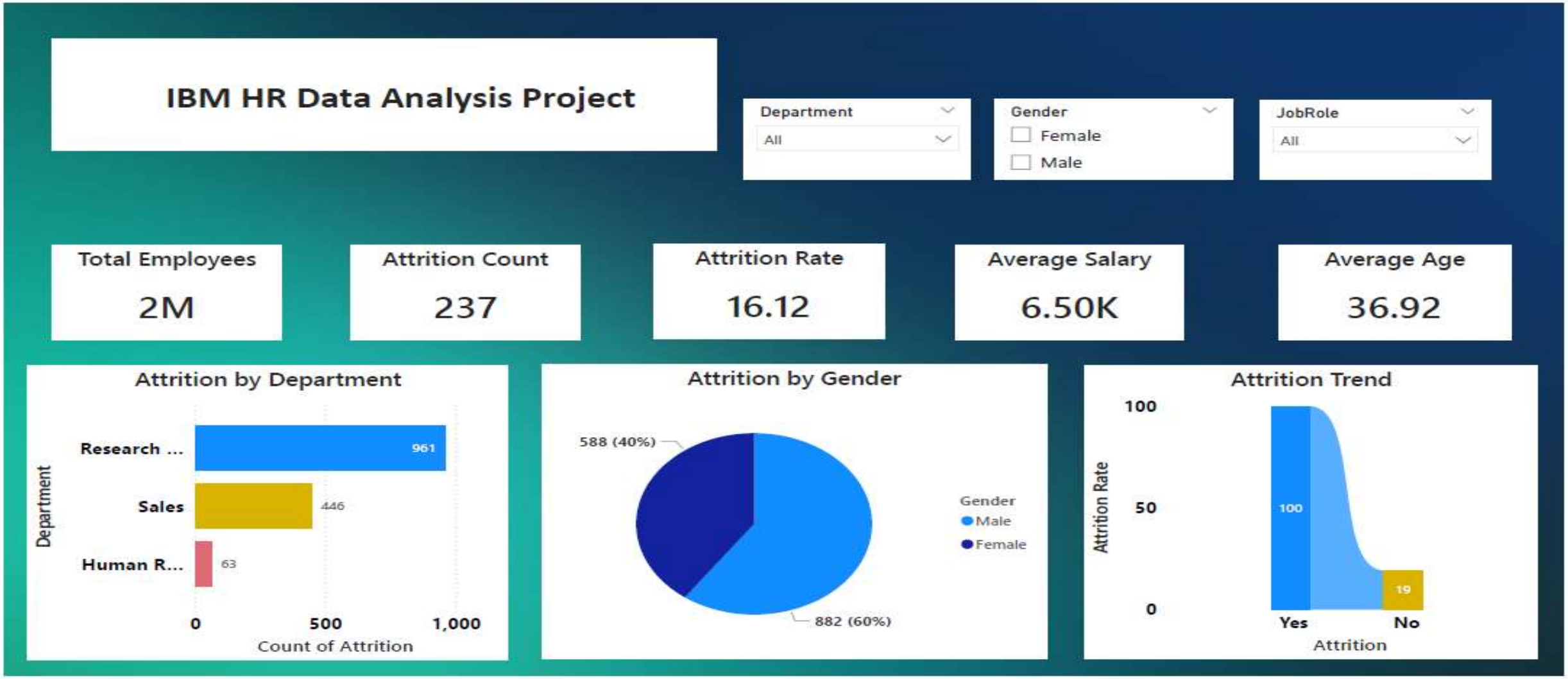
KEY INSIGHTS OF POWER BI ANALYSIS

- 📌 **Education Background**
  → Employees from Life Sciences & Medical fields have the highest attrition rates
- 📌 **Age Group Insights**
  → Majority of attrition occurs in the 26–35 age group (early to mid-career professionals)

KEY INSIGHTS OF POWER BI ANALYSIS

- 📌 Employee Satisfaction & Work-Life Balance
  → Lower satisfaction levels do not strongly correlate with attrition
  → Work-life balance impact is moderate but not dominant
- 📌 Risk Analysis
  ⚠ Identified 12 employees at high risk of attrition

DASHBOARD OF POWER BI ANALYSIS



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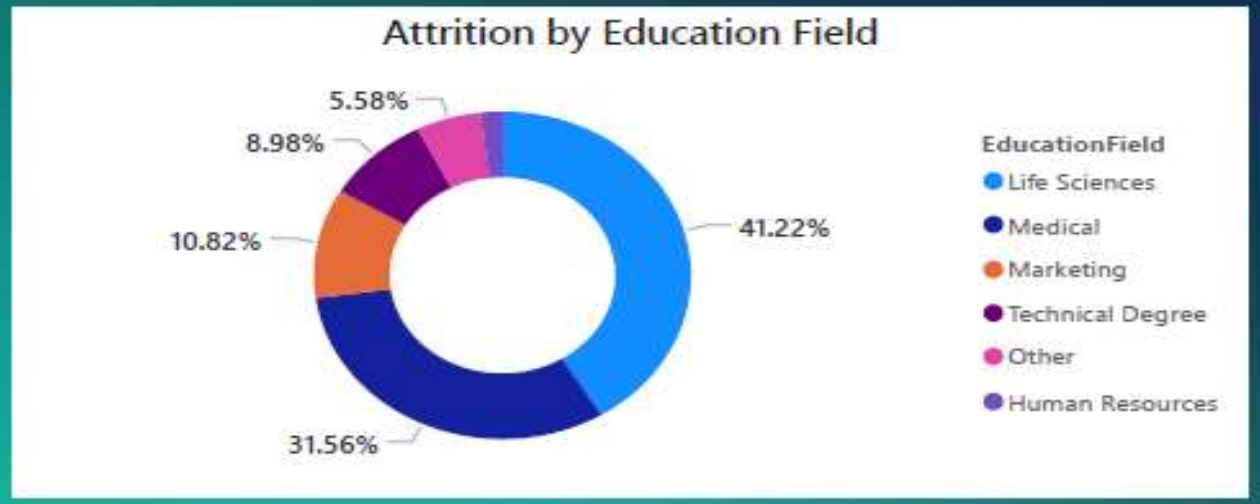

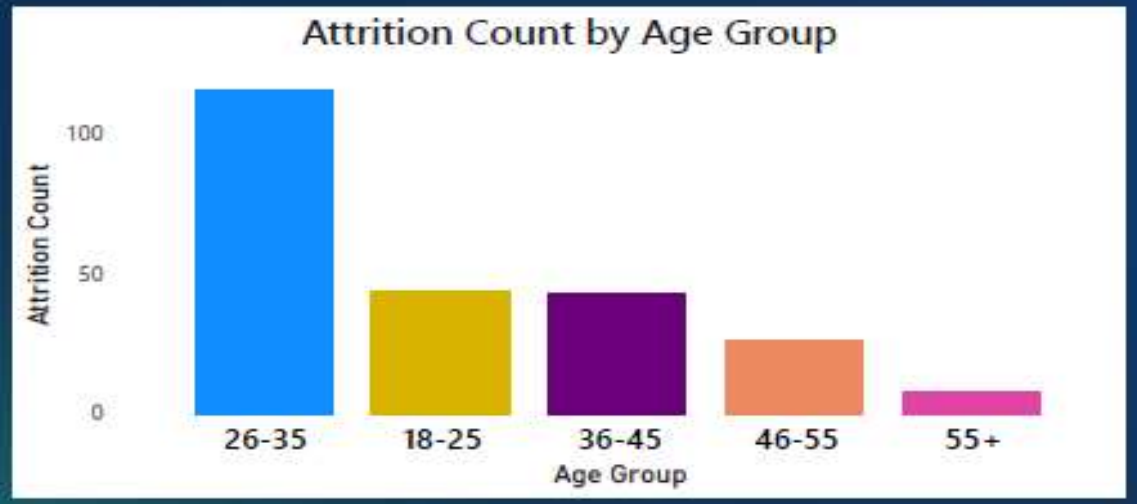

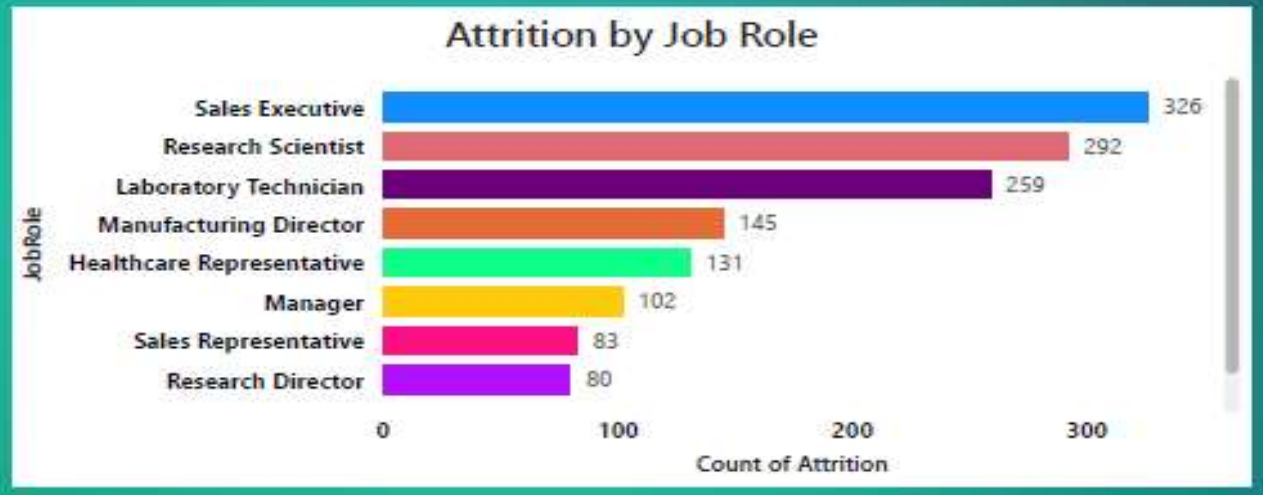

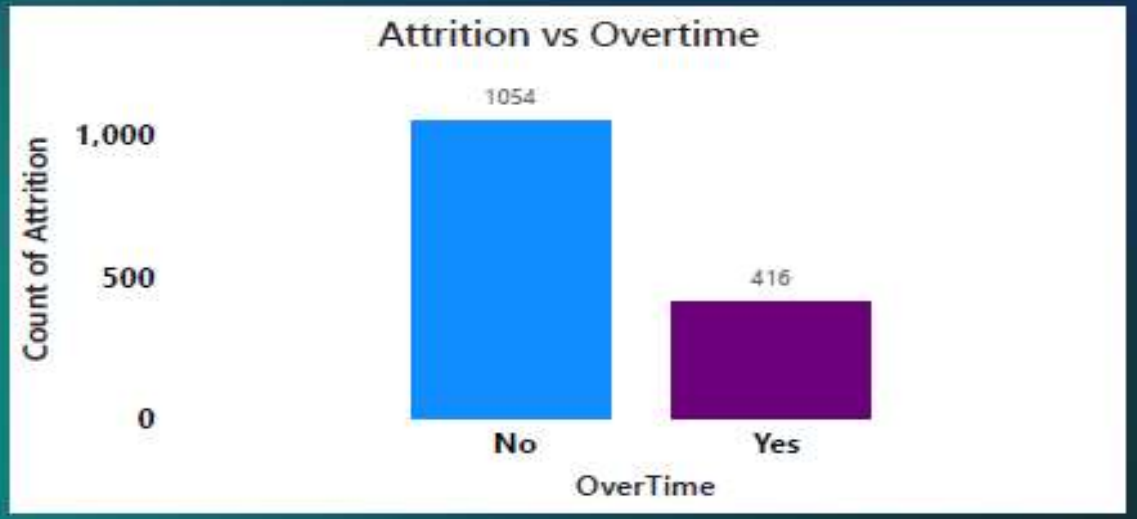

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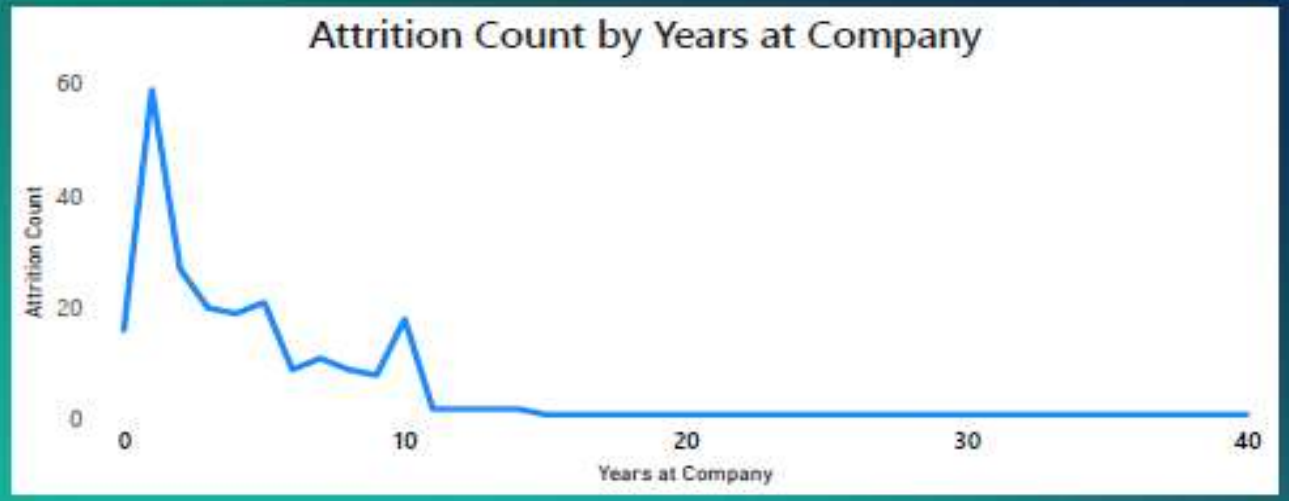

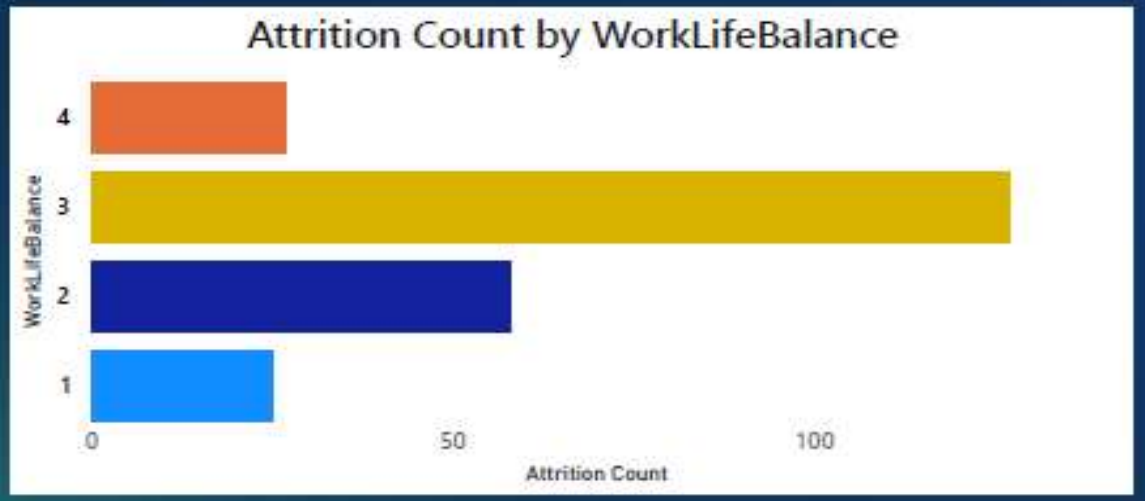

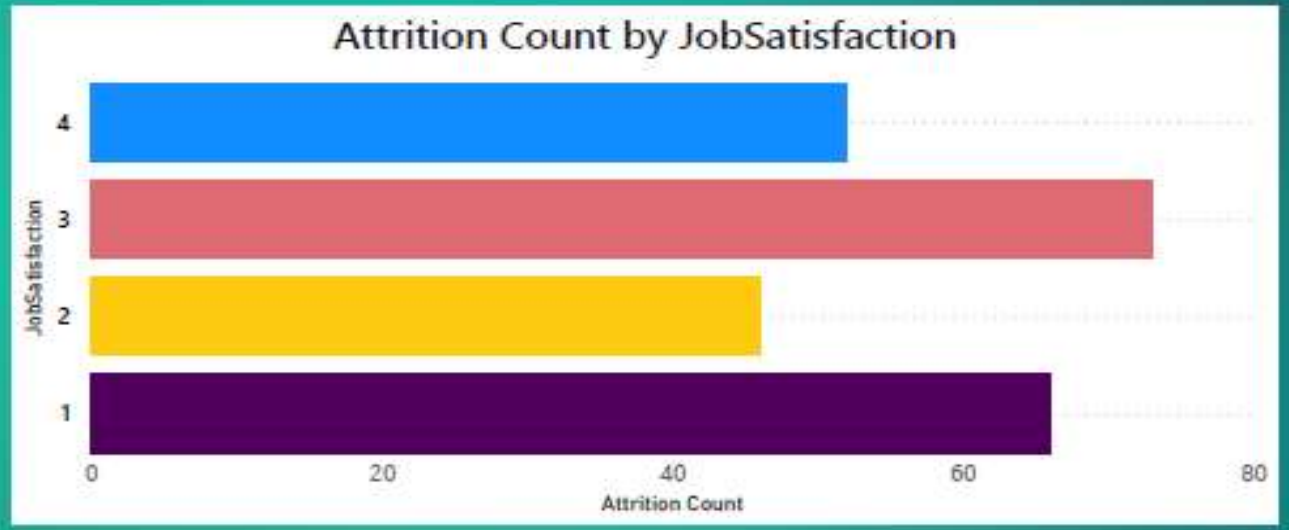

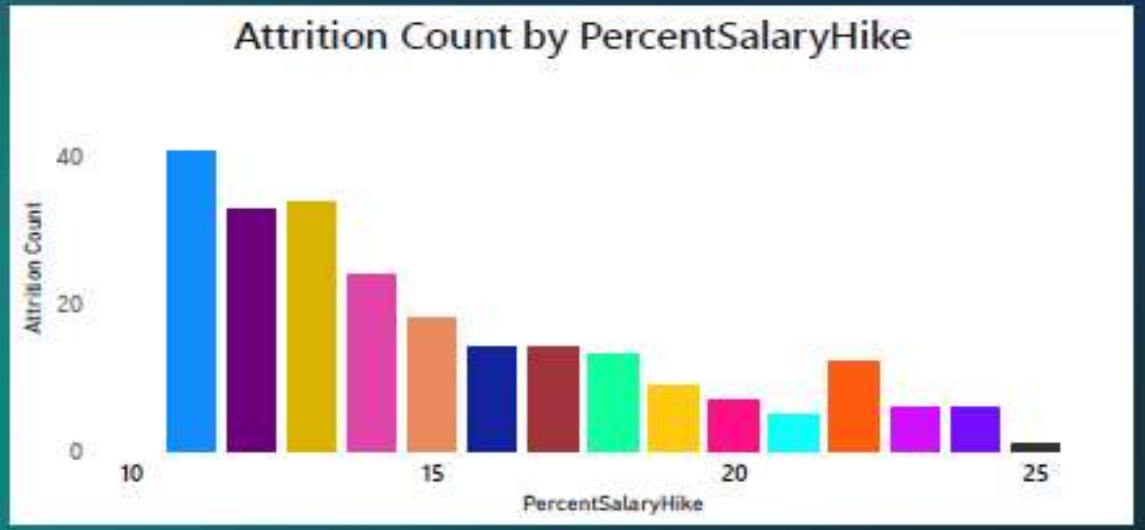

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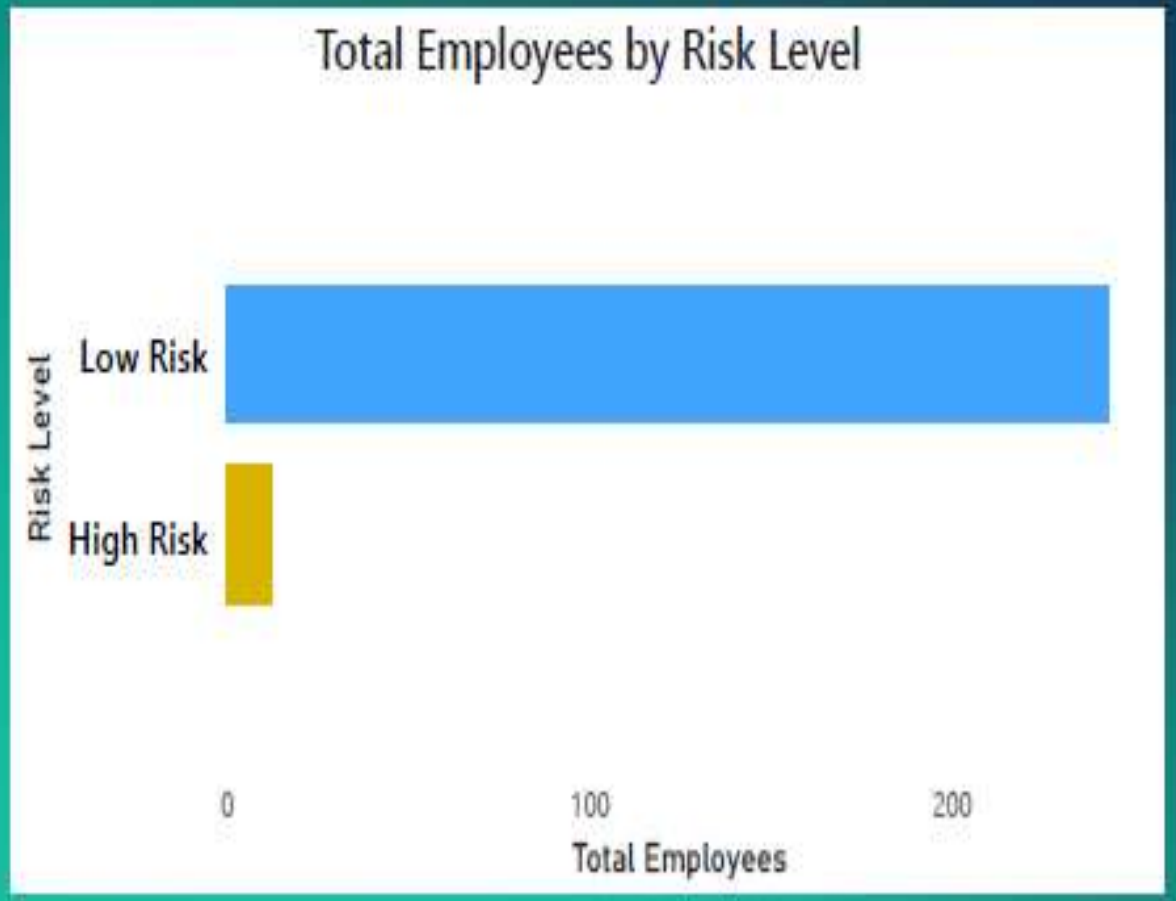

Key Insights of this Dashboard

Research and Development has highest attrition.

Overtime does not increase attrition.

Low Satisfaction does not lead to attrition.

12 Employees are at risk.

CONCLUSION OF THE PROJECT

Performed **data cleaning and preprocessing** on HR dataset using Python (Pandas, NumPy)
Conducted **exploratory data analysis (EDA)** to identify key drivers of employee attrition
Implemented **one-hot encoding and feature engineering** to improve model performance

CONCLUSION OF THE PROJECT

Built and evaluated Logistic Regression and Random Forest models

Achieved improved prediction accuracy using ensemble learning

Identified important features such as overtime, job satisfaction, and tenure

Created visualizations using Matplotlib and Seaborn to present insights